Sagar Varma

Computer Engineering at UC Santa Barbara. I build AI agents and the systems that check them.

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EXPERIENCE

SemiAnalysis Oct 2025 – Sep 2026

Machine Learning Engineer, Remote

- Built deep research AI agents that mapped every semiconductor fab in the world, down to site-level details
- Researched fabs by hand to build ground-truth data, then automated it into LLM evaluations (evals)
- Designed verifier agents that catch hallucinations before they reach the dataset
- Built internal AI tools to visualize and query research data

Avora AI (Y Combinator) Jun 2025 – Sep 2025

Software Engineering Intern, San Francisco, CA

- Scaled the FastAPI backend to 1,000+ dental practices using the product concurrently
- Shipped practice telemetry and quality-of-life features that raised retention 55%

Technologies: Python, TypeScript, React, Remix, PostgreSQL, FastAPI, SQLAlchemy, Kubernetes, GCP BigQuery, AssemblyAI

Intel Jun 2022 – Sep 2022

Software Engineering Intern, Folsom, CA

- Automated manual code updates across millions of lines, cutting merge conflicts 99%
- Shipped security patches for legacy graphics drivers

Technologies: Python, C, Perl, Ruby

PROJECTS

Bardie | textbardie.com

AI assistant that texts college students before they miss deadlines

- Built the agent harness and Canvas LMS connectors; launching with paid plans

Appetite Pal: GLP-1 Food Log | iOS App Store

AI food log for people on GLP-1 medications

- Built and shipped the iOS app; AI tracks calories and every macronutrient
- Sells subscriptions at $14.99/month or $169.99/year; marketed in person at UCSB and to older adults in Santa Barbara

American Time Use Explorer | americantimeuse.com

What Americans do, minute by minute

- Turned the Bureau of Labor Statistics' American Time Use Survey into interactive visualizations

AI Job Application Agent | gettingmejob.com

Computer-use agent that applies to jobs from a sandboxed browser

Earlier projects: wheretowatch.fun, Wherely (~1,500 MAU), AI Test Prep Academy (~1,000 waitlist), 10 Week Shred

TECHNICAL SKILLS

AI Engineering: Large language models (LLMs), generative AI, agentic AI, multi-agent systems, agent harnesses, tool calling, Model Context Protocol (MCP), LLM evals, LLM-as-a-judge, retrieval-augmented generation (RAG), context and prompt engineering

Machine Learning: PyTorch, TensorFlow, fine-tuning, computer vision, natural language processing (NLP)

Frameworks: Claude Agent SDK, OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, Claude and OpenAI APIs, Claude Code

Languages: Python, TypeScript, JavaScript, Swift, Java, C, C++, SQL

Web & Mobile: FastAPI, Django, Flask, Node.js, React (React.js), Remix, SQLAlchemy, Pydantic, REST APIs, OAuth 2.0, SwiftUI (iOS)

Infrastructure: PostgreSQL, Redis, vector databases, BigQuery, Docker, Kubernetes, Amazon Web Services (AWS), Google Cloud Platform (GCP), GitHub Actions (CI/CD)

Hardware: Verilog, VLSI design, Cadence Virtuoso, Cadence SKILL, electronic design automation (EDA), computer architecture

EDUCATION

University of California, Santa Barbara (UCSB) Expected Jun 2027

Bachelor of Science (B.S.), Computer Engineering

- Coursework: AI in Chip Design, VLSI Design, Computer Architecture, Compilers, Semiconductor Fabrication Technology

ADDITIONAL EXPERIENCE

Independent AI Consulting, agents and internal tools for companies (launching) 2026 – Present

Perplexity, Campus Growth Ambassador at UCSB Sep 2025 – Dec 2025